Sonder - database tables

103 tables, 619 columns, grouped by tier. PK = primary key. FK -> table.column shows the table and column a foreign key points to. All tables live in the single **sonder** schema.

Tier 1 - Core: content, diagnostic runs, users and auth, classes, teacher review (43 tables)

app_user
- id (PK)
- email
- role
- display_name
- locale
- time_zone
- status
- created_at

password_credential
- id (PK)
- user_id (FK -> app_user.id)
- password_hash
- changed_at

auth_session
- id (PK)
- user_id (FK -> app_user.id)
- token_hash
- created_at
- expires_at
- revoked_at

auth_token
- id (PK)
- user_id (FK -> app_user.id)
- kind
- token_hash
- created_at
- expires_at
- consumed_at

guardian_link
- id (PK)
- parent_id (FK -> app_user.id)
- student_id (FK -> app_user.id)
- status
- linked_at

classroom
- id (PK)
- name
- note
- join_code
- grade_level
- time_zone
- status
- created_at

class_teacher
- id (PK)
- class_id (FK -> classroom.id)
- teacher_id (FK -> app_user.id)
- status
- assigned_at

enrolment
- id (PK)
- class_id (FK -> classroom.id)
- student_id (FK -> app_user.id)
- status
- joined_at

student_home_class
- id (PK)
- enrolment_id (FK -> enrolment.id)

subject
- id (PK)
- code
- name

class_subject
- id (PK)
- class_id (FK -> classroom.id)
- subject_id (FK -> subject.id)

content_version
- id (PK)
- subject_id (FK -> subject.id)
- version_code
- note
- created_at
- status

topic
- id (PK)
- subject_id (FK -> subject.id)
- external_id

topic_revision
- id (PK)
- topic_id (FK -> topic.id)
- content_version_id (FK -> content_version.id)
- name
- blurb
- grade8_in_scope

misconception
- id (PK)
- subject_id (FK -> subject.id)
- code
- external_id

misconception_revision
- id (PK)
- misconception_id (FK -> misconception.id)
- content_version_id (FK -> content_version.id)
- name
- description
- cognitive_origin
- status
- engine_tier

topic_misconception
- id (PK)
- topic_revision_id (FK -> topic_revision.id)
- misconception_revision_id (FK -> misconception_revision.id)
- target_minimum
- old_topic_label
- review_note
- disposition

source_reference
- id (PK)
- citation_key
- citation
- url
- licence
- scope_note

misconception_source
- id (PK)
- misconception_revision_id (FK -> misconception_revision.id)
- source_id (FK -> source_reference.id)
- locator

item
id (PK)
topic_revision_id (FK -> topic_revision.id)
target_misconception_revision_id (FK -> misconception_revision.id)
external_id
stem
batch
difficulty_est
origin
review_status

item_option
id (PK)
item_id (FK -> item.id)
label
external_id
option_text
is_correct

option_misconception
id (PK)
option_id (FK -> item_option.id)
misconception_revision_id (FK -> misconception_revision.id)
rationale

item_numeric_parameter
id (PK)
item_id (FK -> item.id)
name
value
unit

item_source
id (PK)
item_id (FK -> item.id)
source_id (FK -> source_reference.id)
locator

item_approval
id (PK)
item_id (FK -> item.id)
reviewer_id (FK -> app_user.id)
approved_at
note

hypothesis
id (PK)
content_version_id (FK -> content_version.id)
kind
misconception_revision_id (FK -> misconception_revision.id)

item_prediction
id (PK)
hypothesis_id (FK -> hypothesis.id)
option_id (FK -> item_option.id)
rationale

engine_release
id (PK)
release_code
code_hash
confidence_threshold
minimum_answers
tie_epsilon
eig_floor
unmodelled_threshold
selection_policy
created_at

response_model
id (PK)
content_version_id (FK -> content_version.id)
model_code
source
calibration_note
created_at
status

response_parameter
id (PK)
response_model_id (FK -> response_model.id)
hypothesis_id (FK -> hypothesis.id)
slip
guess

response_probability
id (PK)
response_model_id (FK -> response_model.id)
hypothesis_id (FK -> hypothesis.id)
option_id (FK -> item_option.id)
probability

diagnostic_session
id (PK)
enrolment_id (FK -> enrolment.id)
response_model_id (FK -> response_model.id)
engine_release_id (FK -> engine_release.id)
topic_revision_id (FK -> topic_revision.id)
question_budget
request_key
started_at
ended_at
status

session_hypothesis
id (PK)
session_id (FK -> diagnostic_session.id)
hypothesis_id (FK -> hypothesis.id)
prior_probability

session_question
id (PK)
session_id (FK -> diagnostic_session.id)
step_index
item_id (FK -> item.id)
presented_at
selection_reason
entropy_before
expected_ig

session_step
id (PK)
session_id (FK -> diagnostic_session.id)
step_index
session_question_id (FK -> session_question.id)
answer_option_id (FK -> item_option.id)
answered_at
request_key
calculation_ms

step_posterior
id (PK)
step_id (FK -> session_step.id)
session_hypothesis_id (FK -> session_hypothesis.id)
probability

diagnosis
id (PK)
session_id (FK -> diagnostic_session.id)
created_at

diagnosis_candidate
id (PK)
diagnosis_id (FK -> diagnosis.id)
session_hypothesis_id (FK -> session_hypothesis.id)
rank

escalation
id (PK)
session_id (FK -> diagnostic_session.id)
reason
created_at

escalation_candidate
id (PK)
escalation_id (FK -> escalation.id)
session_hypothesis_id (FK -> session_hypothesis.id)
rank

teacher_review
id (PK)
session_id (FK -> diagnostic_session.id)
revision
teacher_id (FK -> app_user.id)
decision
conclusion
misconception_revision_id (FK -> misconception_revision.id)
note
request_key
decided_at

audit_event
id (PK)
actor_id (FK -> app_user.id)
event_type
relation_name
row_id
request_key
occurred_at
detail

privacy_request
id (PK)
user_id (FK -> app_user.id)
kind
status
requested_at
completed_at
note

Tier 2 - Learning surface (19 tables)

study_note
id (PK)
misconception_revision_id (FK -> misconception_revision.id)
revision
what_was_going_on
how_to_think
example_prompt
wrong_move
right_move
example_answer
origin

study_note_approval
id (PK)
study_note_id (FK -> study_note.id)
teacher_id (FK -> app_user.id)
approved_at
note

intervention
id (PK)
misconception_revision_id (FK -> misconception_revision.id)
revision
script
worked_example

intervention_approval
id (PK)
intervention_id (FK -> intervention.id)
teacher_id (FK -> app_user.id)
approved_at
note

resource
id (PK)
url
title
kind
source

resource_recommendation
id (PK)
resource_id (FK -> resource.id)
misconception_revision_id (FK -> misconception_revision.id)
revision
rationale
relevance
origin

resource_approval
id (PK)
recommendation_id (FK -> resource_recommendation.id)
teacher_id (FK -> app_user.id)
approved_at
note

concept_doc
id (PK)
topic_revision_id (FK -> topic_revision.id)
document_key
revision
title
body
doc_type

concept_doc_misconception
id (PK)
concept_doc_id (FK -> concept_doc.id)
misconception_revision_id (FK -> misconception_revision.id)

concept_doc_chunk
id (PK)
concept_doc_id (FK -> concept_doc.id)
chunk_index
chunk_text

concept_doc_approval
id (PK)
concept_doc_id (FK -> concept_doc.id)
teacher_id (FK -> app_user.id)
approved_at
note

remediation_delivery
id (PK)
review_id (FK -> teacher_review.id)
intervention_id (FK -> intervention.id)
study_note_id (FK -> study_note.id)
status
delivered_at
read_at
completed_at

remediation_resource
id (PK)
delivery_id (FK -> remediation_delivery.id)
recommendation_id (FK -> resource_recommendation.id)

verification_check
id (PK)
delivery_id (FK -> remediation_delivery.id)
kind
scheduled_for

verification_presentation
id (PK)
check_id (FK -> verification_check.id)
item_id (FK -> item.id)
presented_at

verification_result
id (PK)
presentation_id (FK -> verification_presentation.id)
answer_option_id (FK -> item_option.id)
taken_at
result

verification_cancellation
id (PK)
check_id (FK -> verification_check.id)
cancelled_at
reason

parent_summary
id (PK)
review_id (FK -> teacher_review.id)
revision
what_we_found
what_to_work_on
hows_it_going
origin
similarity_score
similarity_floor
created_at

parent_summary_approval
id (PK)
summary_id (FK -> parent_summary.id)
teacher_id (FK -> app_user.id)
approved_at
note

Tier 3 - Messaging and announcements (4 tables)

announcement
id (PK)
class_id (FK -> classroom.id)
author_id (FK -> app_user.id)
title
body
request_key
created_at

dm_thread
id (PK)
enrolment_id (FK -> enrolment.id)
teacher_id (FK -> app_user.id)
guardian_link_id (FK -> guardian_link.id)
created_at

dm_message
id (PK)
thread_id (FK -> dm_thread.id)
sender_id (FK -> app_user.id)
body
sent_at
request_key

dm_read_cursor
id (PK)
thread_id (FK -> dm_thread.id)
reader_id (FK -> app_user.id)
last_read_message_id (FK -> dm_message.id)
read_at

Tier 4 - AI pipeline, trust and evaluation results (35 tables)

ai_model
id (PK)
name
revision
kind
runtime

llm_call_log
id (PK)
model_id (FK -> ai_model.id)
feature
user_id (FK -> app_user.id)
session_id (FK -> diagnostic_session.id)
prompt_hash
prompt_template_revision
prompt_tokens
completion_tokens
latency_ms
outcome
request_key
created_at
payload

cohort_observation
id (PK)
class_id (FK -> classroom.id)
subject_id (FK -> subject.id)
definition
window_start
window_end
sample_size
value
unit
query_revision
observed_at

evidence_reference
id (PK)
step_id (FK -> session_step.id)
verification_id (FK -> verification_result.id)
review_id (FK -> teacher_review.id)
cohort_observation_id (FK -> cohort_observation.id)
item_id (FK -> item.id)
chunk_id (FK -> concept_doc_chunk.id)
source_id (FK -> source_reference.id)
external_url

llm_grounding
id (PK)
call_id (FK -> llm_call_log.id)
evidence_id (FK -> evidence_reference.id)
position

assistant_thread
id (PK)
owner_id (FK -> app_user.id)
kind
misconception_revision_id (FK -> misconception_revision.id)
enrolment_id (FK -> enrolment.id)
class_id (FK -> classroom.id)
escalation_id (FK -> escalation.id)
created_at

assistant_message
- id (PK)
- thread_id (FK -> assistant_thread.id)
- sequence
- speaker
- body
- preset_key
- call_id (FK -> llm_call_log.id)
- outcome
- created_at

assistant_message_evidence
- id (PK)
- message_id (FK -> assistant_message.id)
- evidence_id (FK -> evidence_reference.id)
- position
- display_text

consultant_suggestion
- id (PK)
- call_id (FK -> llm_call_log.id)
- enrolment_id (FK -> enrolment.id)
- escalation_id (FK -> escalation.id)
- message_id (FK -> assistant_message.id)
- claim
- confidence
- created_at

suggestion_evidence
- id (PK)
- suggestion_id (FK -> consultant_suggestion.id)
- evidence_id (FK -> evidence_reference.id)
- position
- display_text

suggestion_review
- id (PK)
- suggestion_id (FK -> consultant_suggestion.id)
- revision
- teacher_id (FK -> app_user.id)
- verdict
- note
- reviewed_at

coverage_gap
- id (PK)
- topic_revision_id (FK -> topic_revision.id)
- kind
- target_minimum
- note
- detected_from_session_id (FK -> diagnostic_session.id)
- detected_at
- bank_count_at_detection
- status
- scan_key

gap_misconception
- id (PK)
- gap_id (FK -> coverage_gap.id)
- misconception_revision_id (FK -> misconception_revision.id)

generation_run
- id (PK)
- gap_id (FK -> coverage_gap.id)
- drafter_model_id (FK -> ai_model.id)
- scorer_model_id (FK -> ai_model.id)
- pass_mark
- target_min
- target_max
- max_rounds
- request_key
- started_at
- status

generation_round
- id (PK)
- run_id (FK -> generation_run.id)
- round_index
- created_at

generated_item
- id (PK)
- round_id (FK -> generation_round.id)
- candidate_index
- item_id (FK -> item.id)
- drafter_call_id (FK -> llm_call_log.id)
- distractor_source
- mapping_source
- status
- rejection_reason
- published_item_id (FK -> item.id)

quality_rubric
- id (PK)
- rubric_code
- definition

rubric_dimension
- id (PK)
- rubric_id (FK -> quality_rubric.id)
- code
- definition

quality_assessment
- id (PK)
- generated_item_id (FK -> generated_item.id)
- rubric_id (FK -> quality_rubric.id)
- scorer_model_id (FK -> ai_model.id)
- judge_call_id (FK -> llm_call_log.id)
- overall
- reason
- created_at

quality_dimension_score
- id (PK)
- assessment_id (FK -> quality_assessment.id)
- dimension_id (FK -> rubric_dimension.id)
- score

content_review_assignment
- id (PK)
- generated_item_id (FK -> generated_item.id)
- reviewer_id (FK -> app_user.id)
- assigned_at

generation_review
- id (PK)
- assignment_id (FK -> content_review_assignment.id)
- verdict
- note
- decided_at

review_dispute
- id (PK)
- generated_item_id (FK -> generated_item.id)
- option_id (FK -> item_option.id)
- issue_key
- issue

reviewer_mapping_proposal
- id (PK)
- review_id (FK -> generation_review.id)
- option_id (FK -> item_option.id)
- misconception_revision_id (FK -> misconception_revision.id)
- rationale

review_adjudication
id (PK)
dispute_id (FK -> review_dispute.id)
adjudicator_id (FK -> app_user.id)
resolution
decided_at

provenance_entry
id (PK)
item_id (FK -> item.id)
distractor_item_id (FK -> item.id)
note_id (FK -> study_note.id)
recommendation_id (FK -> resource_recommendation.id)
suggestion_id (FK -> consultant_suggestion.id)
run_id (FK -> generation_run.id)
summary_id (FK -> parent_summary.id)
doc_id (FK -> concept_doc.id)
message_id (FK -> assistant_message.id)
purpose
note
created_at

provenance_call
id (PK)
provenance_id (FK -> provenance_entry.id)
call_id (FK -> llm_call_log.id)

provenance_grounding
id (PK)
provenance_id (FK -> provenance_entry.id)
evidence_id (FK -> evidence_reference.id)

provenance_validation
id (PK)
provenance_id (FK -> provenance_entry.id)
validator_id (FK -> app_user.id)
decision
note
validated_at

evaluation_study
id (PK)
study_code
definition
dataset_reference
dataset_hash
window_start
window_end
caveat
engine_release_id (FK -> engine_release.id)
content_version_id (FK -> content_version.id)
created_at

evaluation_arm
id (PK)
study_id (FK -> evaluation_study.id)
code
definition

metric_definition
id (PK)
code
revision
label
definition
denominator_definition
unit
good_when

evaluation_result
id (PK)
arm_id (FK -> evaluation_arm.id)
metric_id (FK -> metric_definition.id)
subject_id (FK -> subject.id)
stratum
value
sample_size
numerator
denominator
lower_bound
upper_bound
caveat

evaluation_comparison
id (PK)
result_id (FK -> evaluation_result.id)
baseline_result_id (FK -> evaluation_result.id)
comparison_note

job_run
id (PK)
kind
request_key
status
started_at
ended_at
heartbeat_at
error_text

Tier 5 - Vector tables (pgvector) (2 tables)

item_embedding
id (PK)
item_id (FK -> item.id)
model_id (FK -> ai_model.id)
text_hash
indexed_at
embedding

concept_chunk_embedding
id (PK)
chunk_id (FK -> concept_doc_chunk.id)
model_id (FK -> ai_model.id)
text_hash
indexed_at
embedding